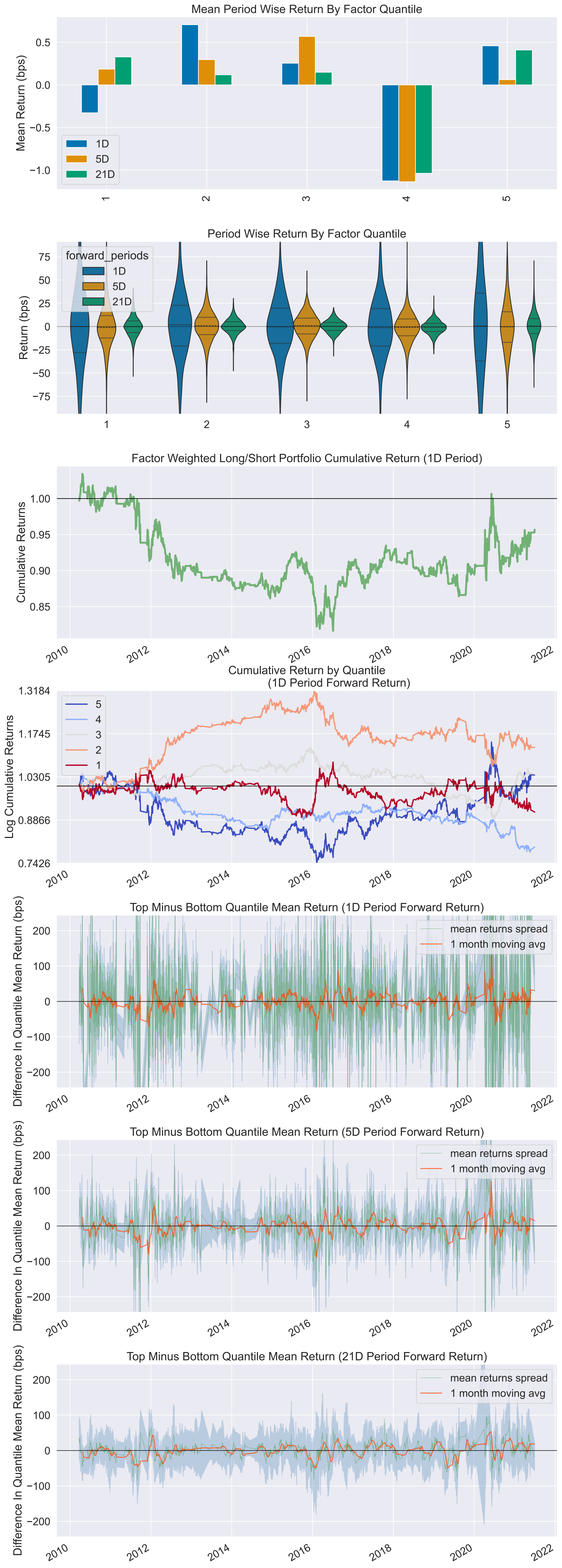
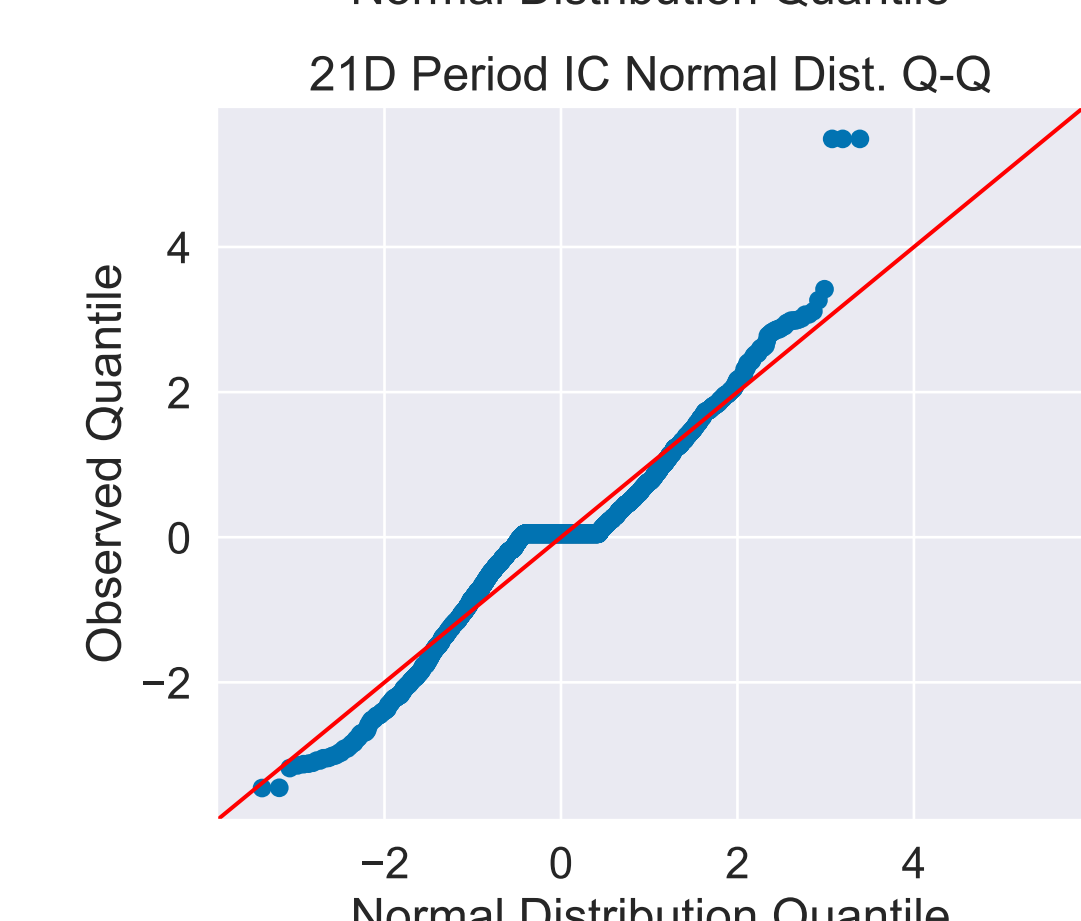
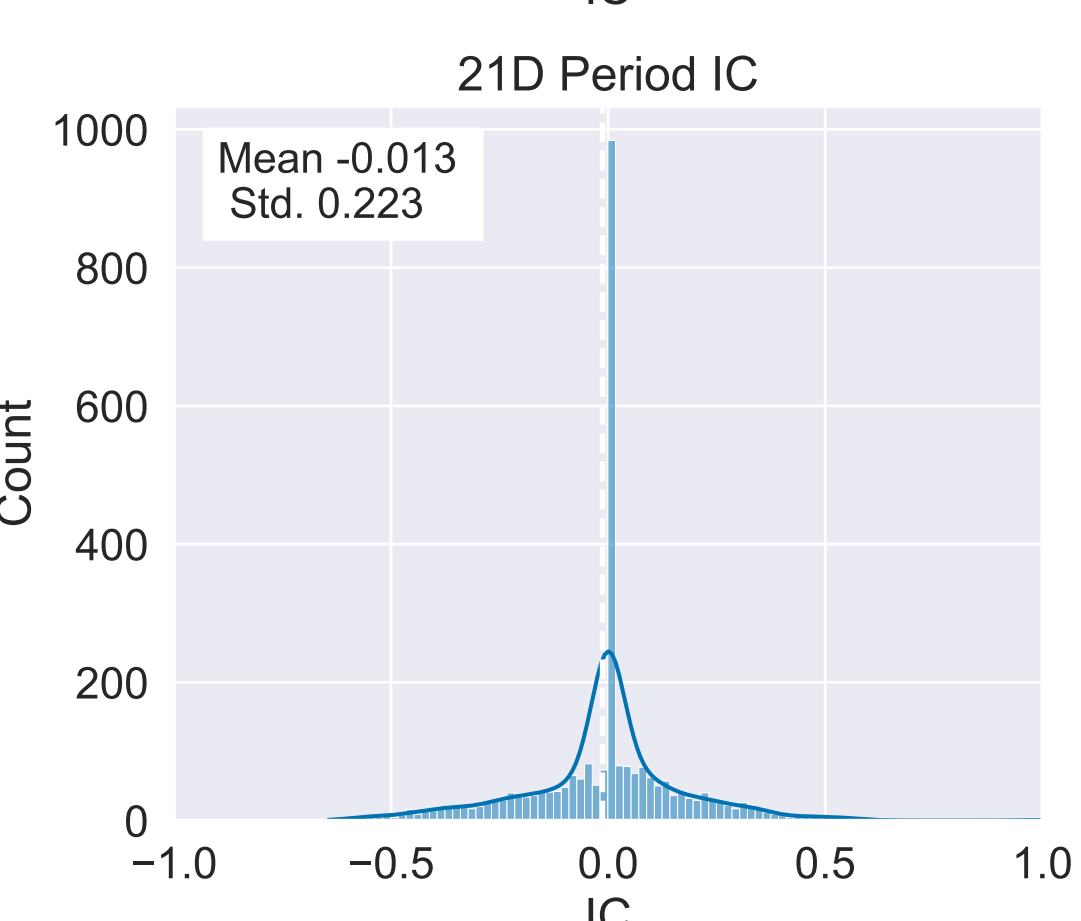
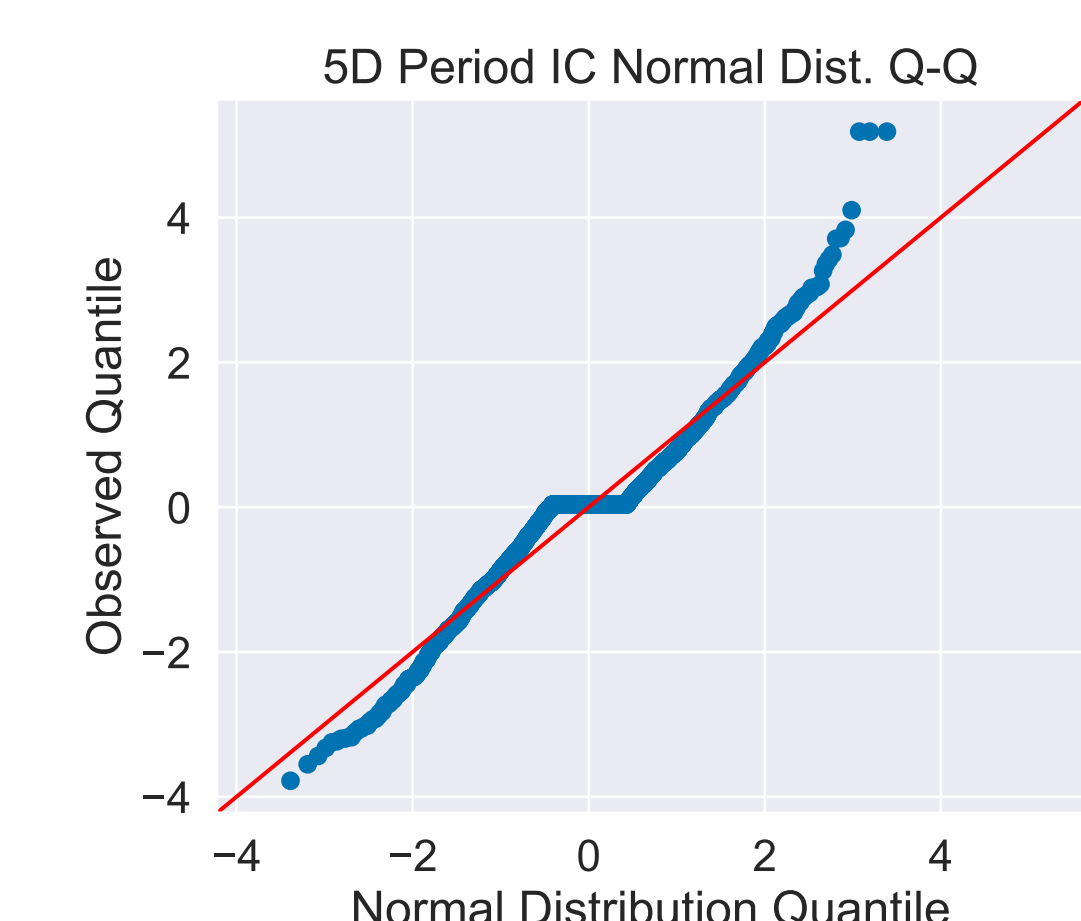
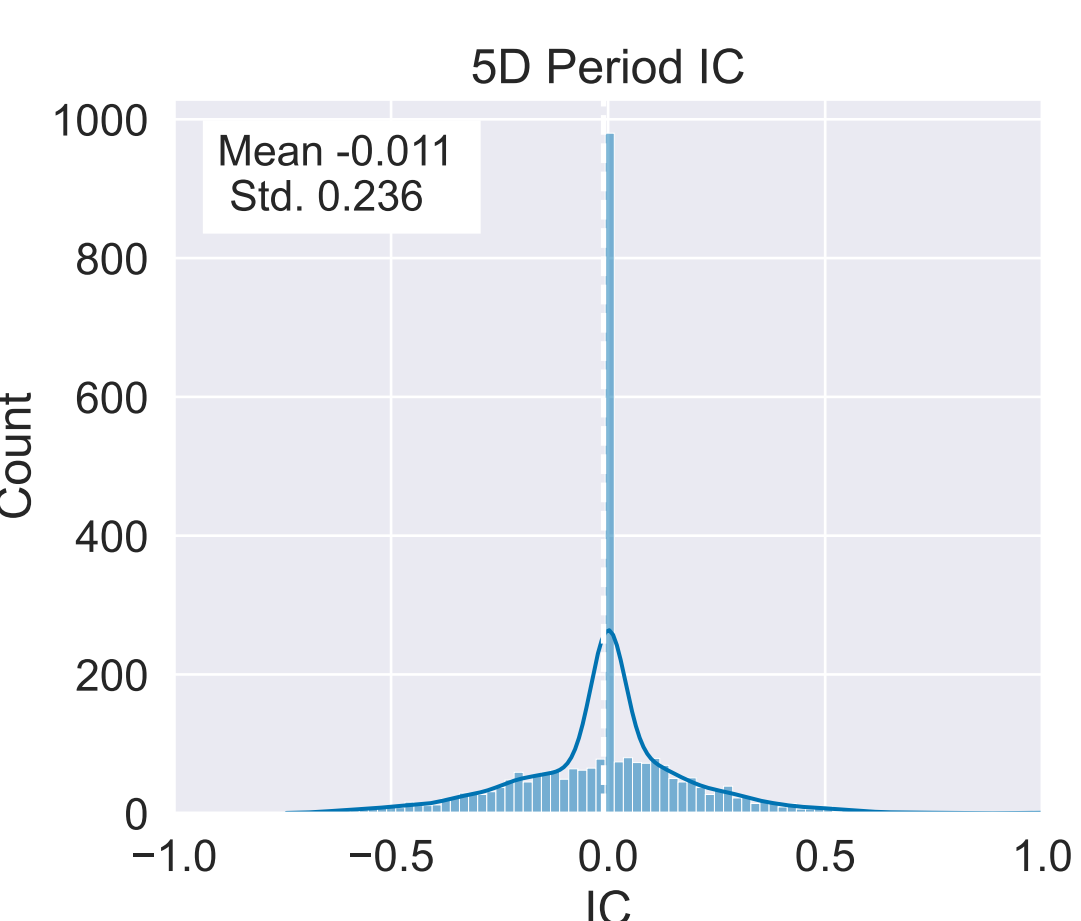
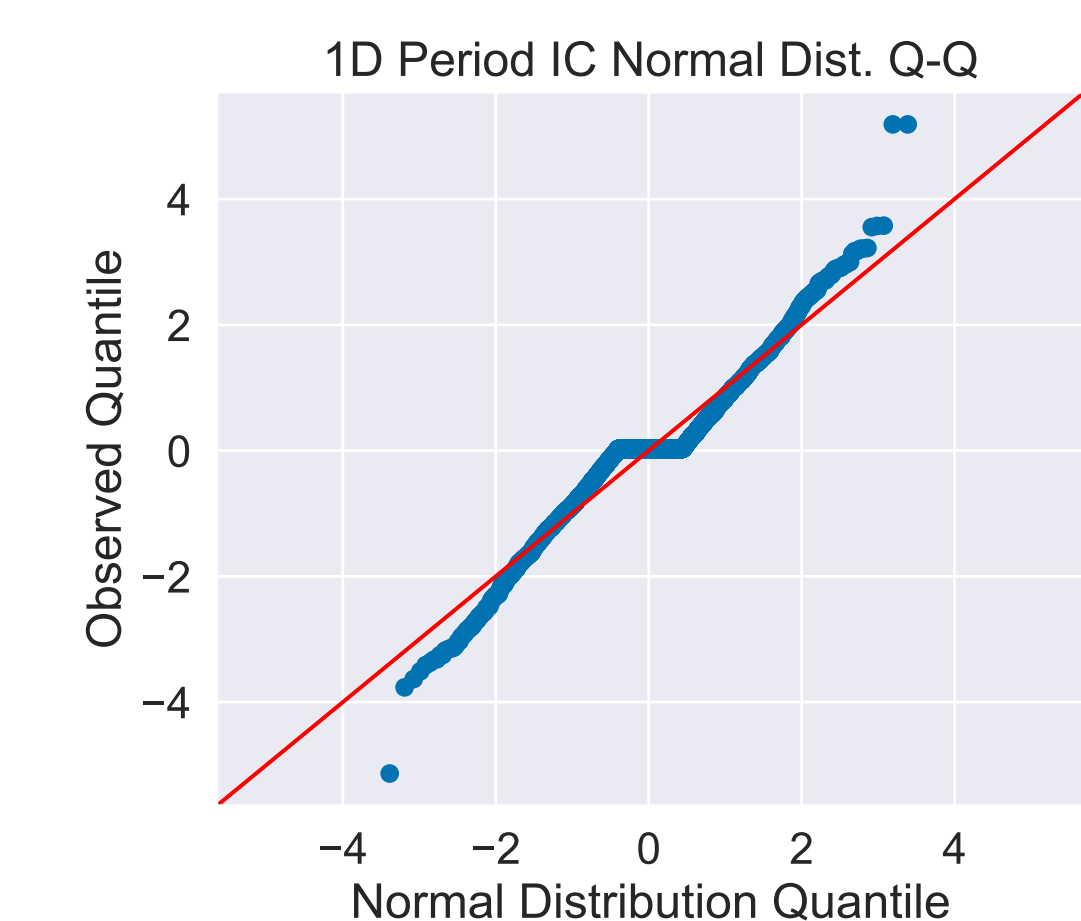
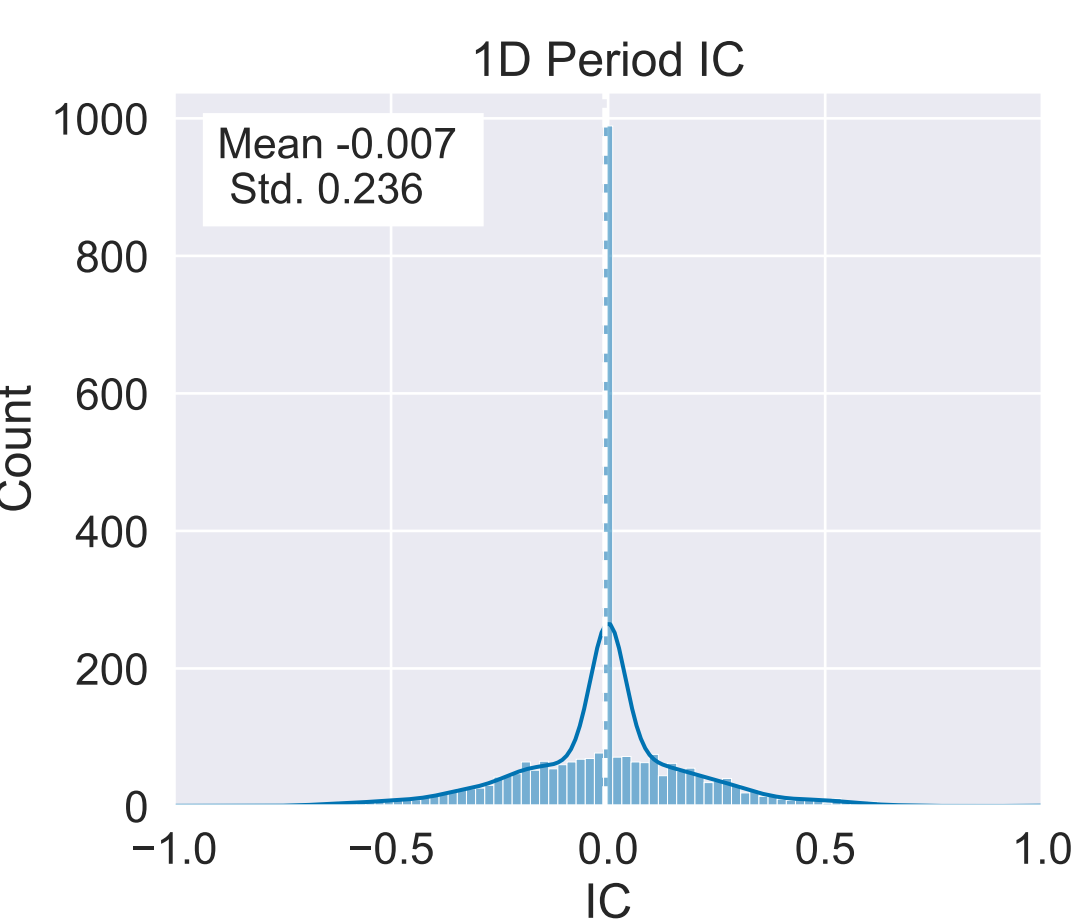
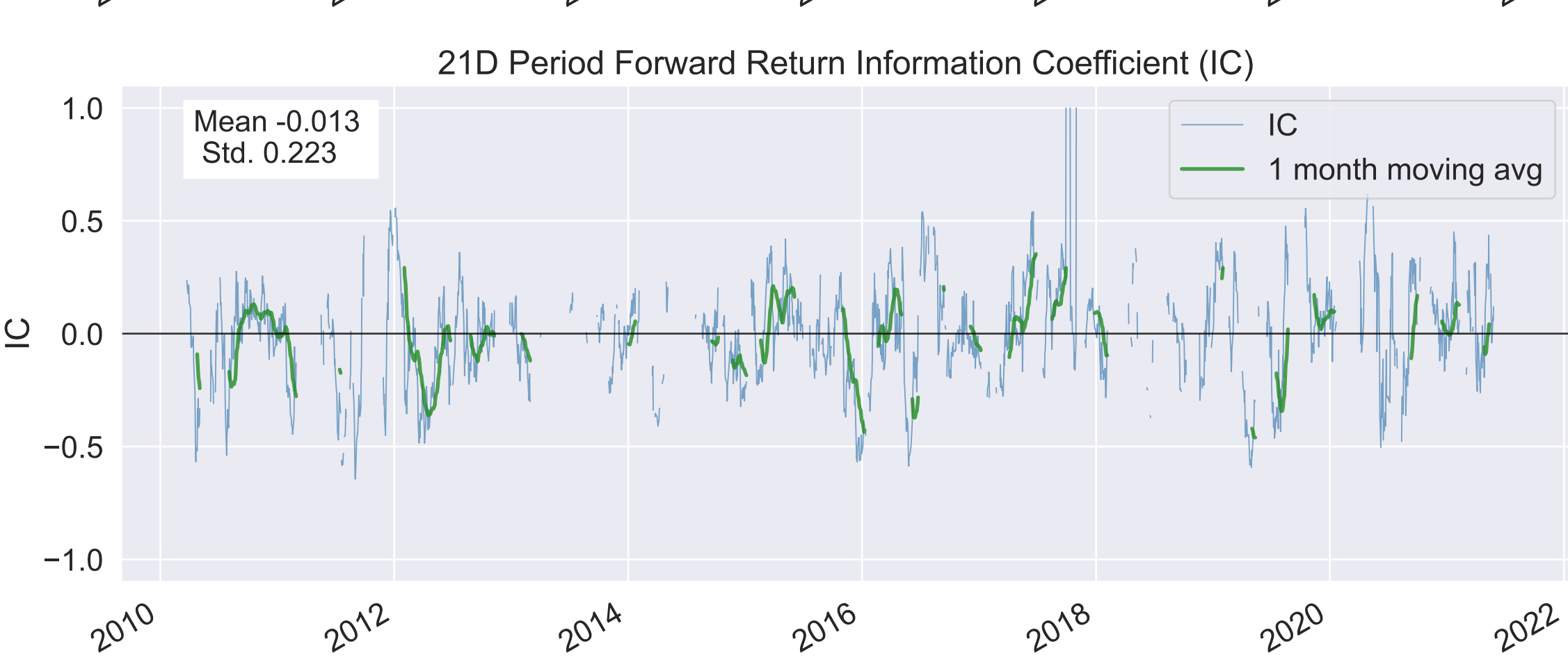
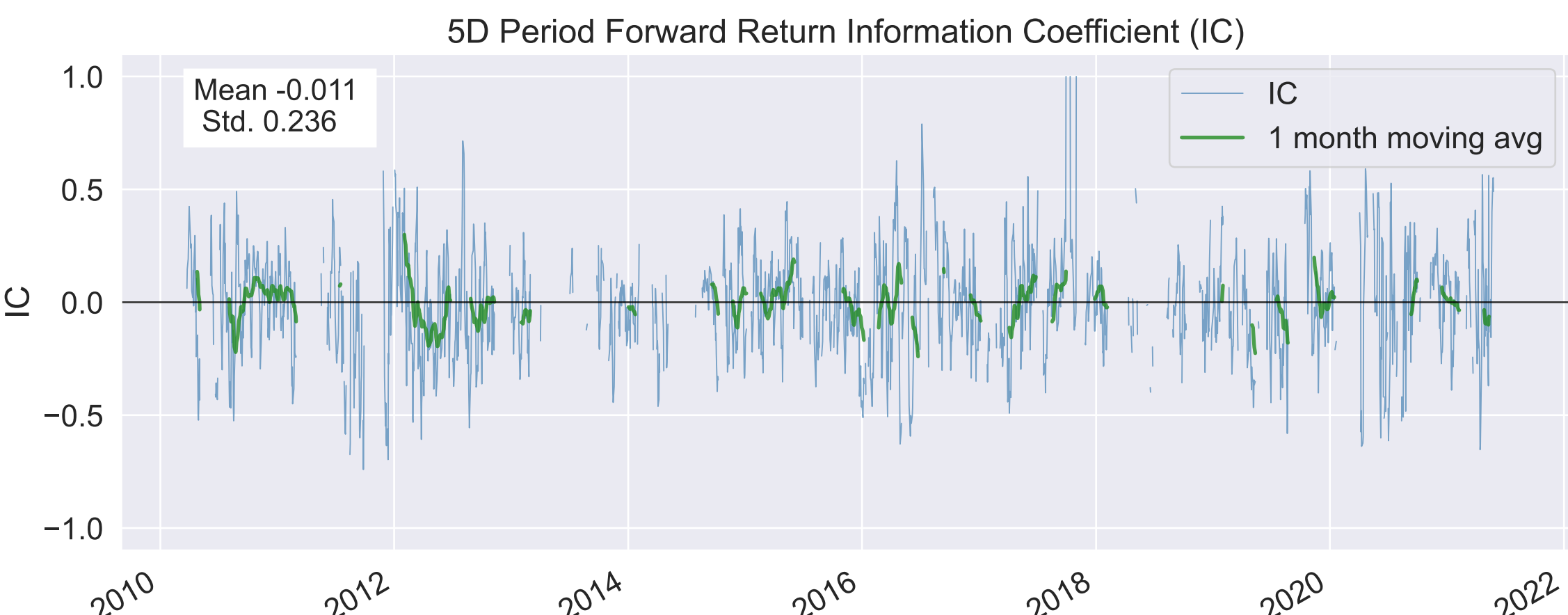
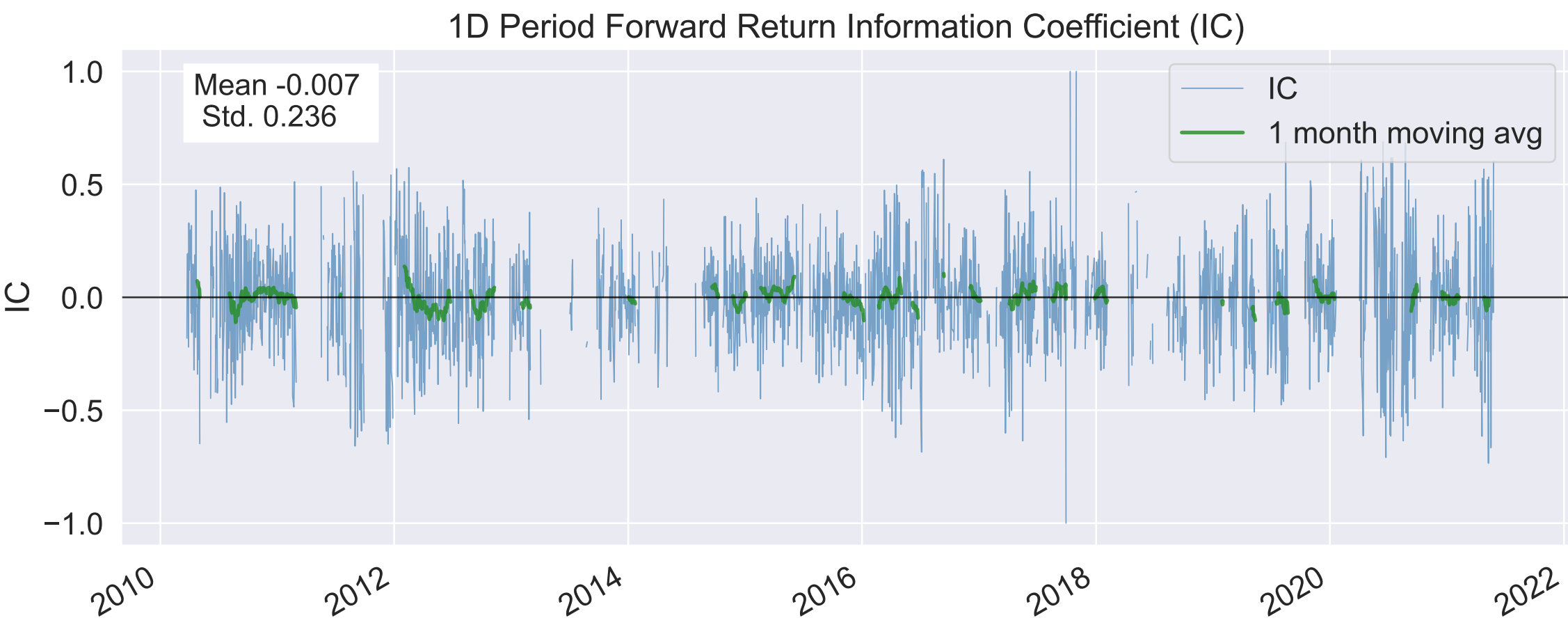






### Monthly Mean 1D Period IC

|      |          |        |         |         |        |          |         |         |         |        |         |         |      |
|------|----------|--------|---------|---------|--------|----------|---------|---------|---------|--------|---------|---------|------|
| 2010 |          |        |         | 0.063   | 0.034  | -0.29    | -0.051  | 0.025   | -0.08   | 0.012  | 0.025   | 0.022   | 0.02 |
| 2011 | -0.00065 | -0.039 | -0.38   |         | 0.086  | 0.0069   | -0.016  | -0.054  | -0.17   |        | 0.2     | -0.14   |      |
| 2012 | 0.12     | 0.0025 | -0.045  | -0.043  | -0.074 | -0.00022 | -0.078  | -0.0027 | -0.0085 | 0.0026 | 0.014   | -0.048  |      |
| 2013 | -0.026   | -0.018 | -0.18   | -0.26   |        | 0.099    | 0.013   | -0.19   | 0.17    | 0.045  | -0.074  | -0.0021 |      |
| 2014 | -0.022   | -0.045 | 0.017   | 0.016   | -0.02  | -0.21    | -0.1    | 0.031   | 0.052   | -0.042 | -0.056  | 0.018   |      |
| 2015 | -0.062   | 0.022  | 0.044   | -0.0093 | 0.092  | 0.028    | -0.089  | -0.047  | -0.0045 | -0.031 | -0.0023 | -0.056  |      |
| 2016 | -0.12    | -0.015 | -0.01   | 0.061   | -0.041 | -0.12    | 0.18    | 0.13    | 0.039   | -0.04  | 0.052   | -0.015  |      |
| 2017 | 0.0084   | -0.11  | -0.0013 | -0.029  | 0.017  | 0.037    | -0.0084 | 0.05    | -0.03   | 0.2    | -0.037  | 0.002   |      |
| 2018 | -0.013   | 0.013  | -0.0078 | -0.034  | 0.33   | -0.15    | -0.079  | -0.026  | -0.055  | -0.065 | 0.013   | 0.02    |      |
| 2019 | -0.028   | 0.0076 | 0.03    | -0.042  | -0.14  | 0.074    | -0.021  | -0.087  |         | 0.074  | 0.024   | -0.0025 |      |
| 2020 | 0.0069   |        |         | 0.032   | 0.039  | -0.016   | -0.096  | -0.16   | 0.054   | 0.049  | -0.016  | 0.0054  |      |
| 2021 | -0.01    | 0.039  | 0.026   | -0.047  | 0.036  | 0.23     |         |         |         |        |         |         |      |
|      | 1        | 2      | 3       | 4       | 5      | 6        | 7       | 8       | 9       | 10     | 11      | 12      |      |

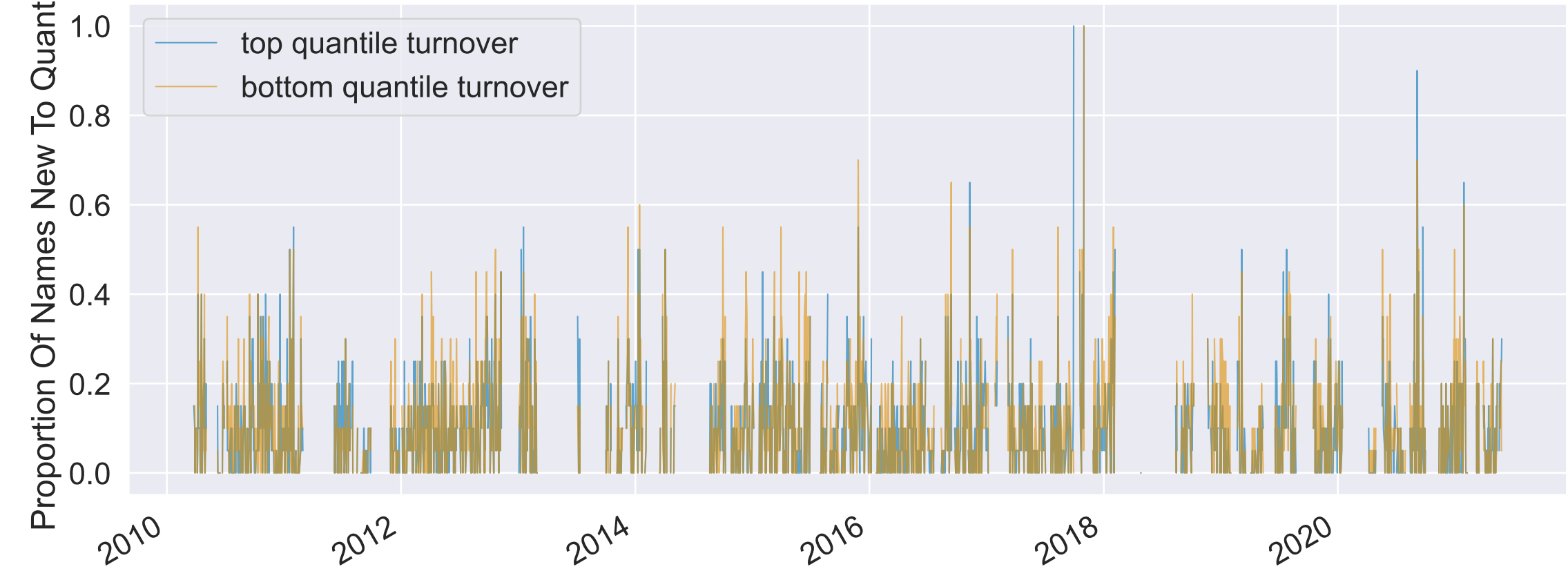
### Monthly Mean 5D Period IC

|      |         |        |        |        |        |        |        |         |        |        |        |        |       |       |
|------|---------|--------|--------|--------|--------|--------|--------|---------|--------|--------|--------|--------|-------|-------|
| 2010 |         |        |        |        | 0.21   | 0.019  | -0.34  | -0.089  | 0.057  | -0.15  | 0.028  | 0.098  | 0.044 | 0.022 |
| 2011 | 0.068   | -0.1   | -0.24  |        | 0.086  | 0.096  | -0.089 | -0.22   | -0.28  |        | 0.46   | -0.19  |       |       |
| 2012 | 0.28    | -0.042 | -0.083 | -0.1   | -0.15  | 0.047  | -0.1   | -0.0056 | -0.021 | 0.006  | -0.084 | 0.22   |       |       |
| 2013 | -0.082  | -0.026 | -0.17  | -0.092 |        | -0.018 | 0.13   | -0.045  | 0.14   | 0.038  | -0.14  | -0.021 |       |       |
| 2014 | -0.047  | 0.17   | -0.05  | -0.19  | -0.22  | -0.051 | -0.039 | 0.07    | 0.049  | -0.04  | -0.088 | 0.044  |       |       |
| 2015 | -0.023  | -0.03  | 0.049  | -0.021 | 0.21   | 0.031  | 0.014  | -0.13   | 0.0069 | 0.025  | -0.053 | -0.11  |       |       |
| 2016 | -0.31   | 0.014  | -0.09  | 0.13   | -0.026 | -0.22  | 0.28   | 0.21    | 0.077  | -0.055 | 0.027  | -0.059 |       |       |
| 2017 | -0.12   | -0.18  | -0.072 | -0.043 | 0.0096 | 0.13   | -0.067 | 0.068   | 0.15   | 0.21   | -0.13  | 0.035  |       |       |
| 2018 | -0.0073 | -0.16  | -0.3   | -0.07  | 0.2    | -0.2   | 0.033  | -0.016  | 0.0056 | -0.013 | 0.026  | -0.046 |       |       |
| 2019 | 0.06    | -0.042 | 0.045  | -0.064 | -0.32  | 0.036  | -0.042 | -0.18   |        | 0.34   | -0.047 | 0.019  |       |       |
| 2020 | 0.054   |        |        |        | 0.07   | 0.2    | -0.1   | -0.083  | -0.14  | 0.091  | 0.014  | 0.15   | 0.014 |       |
| 2021 | -0.029  | 0.05   | 0.082  | -0.1   | 0.12   | -0.38  |        |         |        |        |        |        |       |       |
|      | 1       | 2      | 3      | 4      | 5      | 6      | 7      | 8       | 9      | 10     | 11     | 12     |       |       |

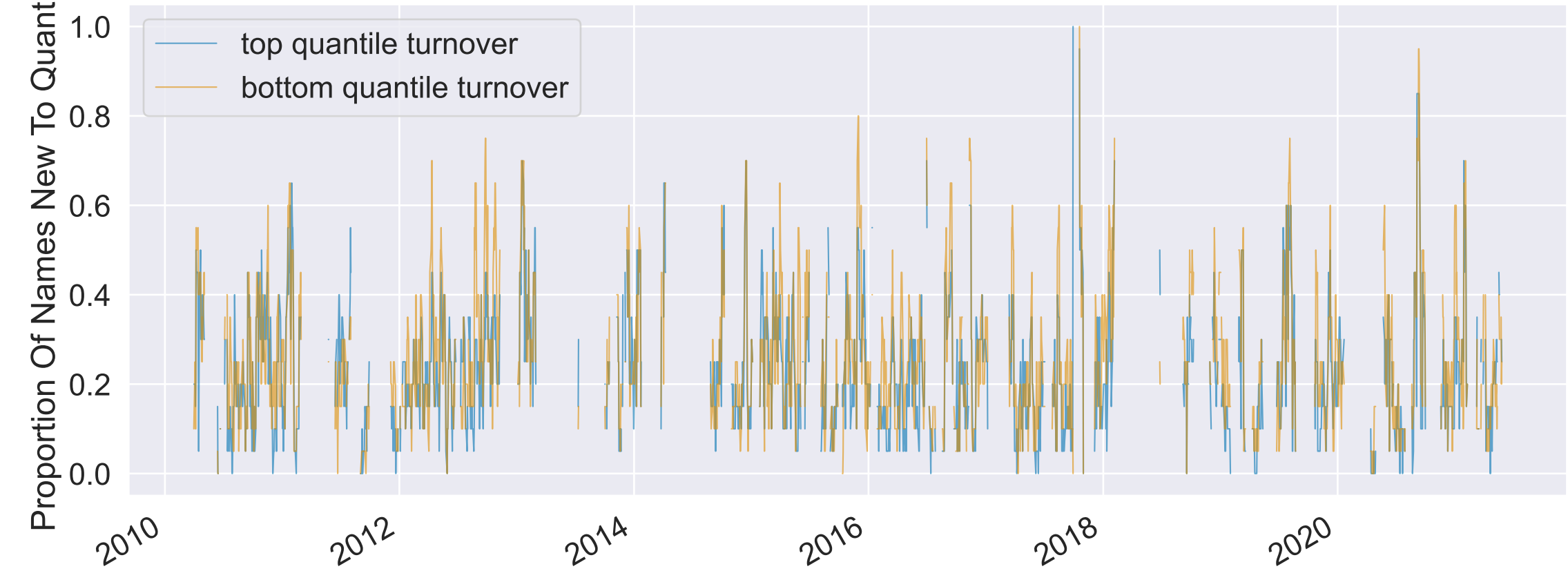
### Monthly Mean 21D Period IC

|      |        |        |       |         |        |        |         |        |        |        |        |         |  |  |
|------|--------|--------|-------|---------|--------|--------|---------|--------|--------|--------|--------|---------|--|--|
| 2010 |        |        |       | 0.21    | -0.21  | -0.37  | -0.12   | -0.034 | 0.092  | 0.093  | 0.098  | -0.0067 |  |  |
| 2011 | 0.022  | -0.3   | -0.13 |         | -0.045 | 0.023  | -0.41   | -0.29  | -0.2   |        | -0.26  | 0.15    |  |  |
| 2012 | 0.32   | -0.13  | -0.24 | -0.34   | -0.065 | -0.042 | 0.083   | -0.049 | -0.055 | 0.0017 | 0.0057 | 0.082   |  |  |
| 2013 | 0.0005 | -0.12  | -0.23 | -0.0072 |        | 0.09   | 0.13    | 0.003  | 0.068  | 0.055  | -0.1   | -0.048  |  |  |
| 2014 | 0.079  | 0.12   | -0.29 | -0.21   | 0.14   | 0.16   | 0.075   | 0.033  | -0.049 | 0.025  | -0.16  | -0.17   |  |  |
| 2015 | 0.089  | -0.16  | 0.2   | 0.07    | 0.2    | 0.056  | -0.03   | -0.041 | -0.085 | 0.12   | -0.2   | -0.38   |  |  |
| 2016 | -0.35  | 0.041  | 0.071 | 0.094   | -0.22  | -0.19  | 0.4     | 0.25   | 0.014  | -0.066 | 0.044  | -0.058  |  |  |
| 2017 | -0.16  | -0.24  | -0.12 | 0.084   | 0.083  | 0.35   | -0.046  | 0.15   | 0.3    | 0.13   | -0.026 | 0.11    |  |  |
| 2018 | -0.08  | -0.22  | -0.35 | 0.12    | 0.19   | -0.18  | -0.0093 | 0.11   | -0.04  | -0.1   | -0.19  | 0.0095  |  |  |
| 2019 | 0.29   | -0.001 | 0.16  | -0.4    | -0.34  | 0.035  | -0.32   | 0.11   |        | 0.31   | 0.019  | 0.1     |  |  |
| 2020 | 0.057  |        |       | 0.27    | 0.28   | -0.24  | -0.032  | -0.24  | 0.19   | 0.24   | 0.13   | 0.0089  |  |  |
| 2021 | 0.14   | 0.033  | 0.1   | -0.097  | 0.18   | -0.24  |         |        |        |        |        |         |  |  |
|      | 1      | 2      | 3     | 4       | 5      | 6      | 7       | 8      | 9      | 10     | 11     | 12      |  |  |

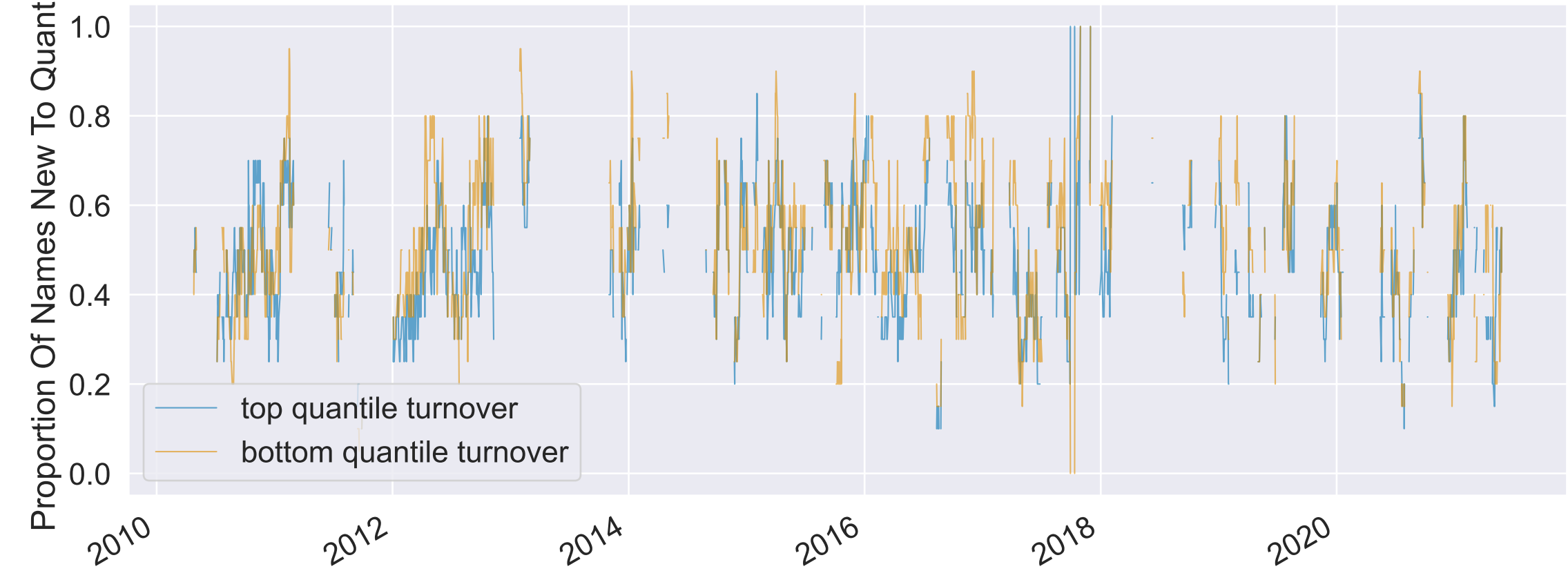
1D Period Top and Bottom Quantile Turnover



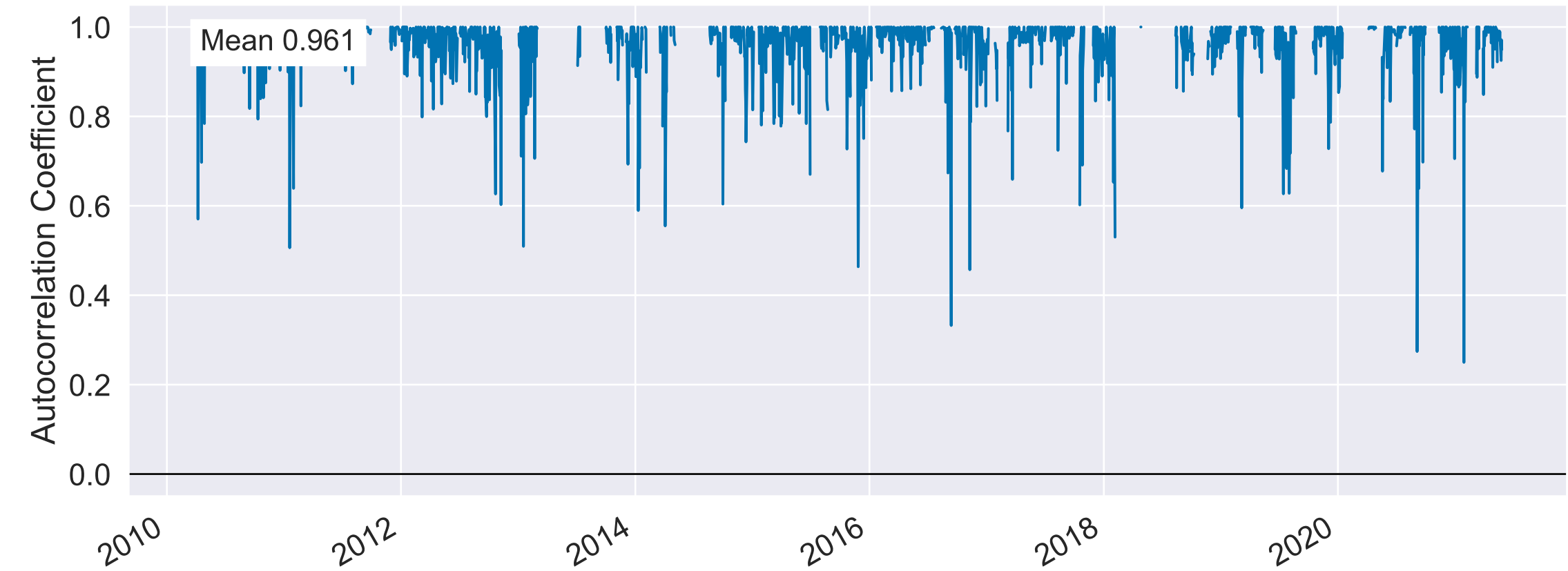
5D Period Top and Bottom Quantile Turnover



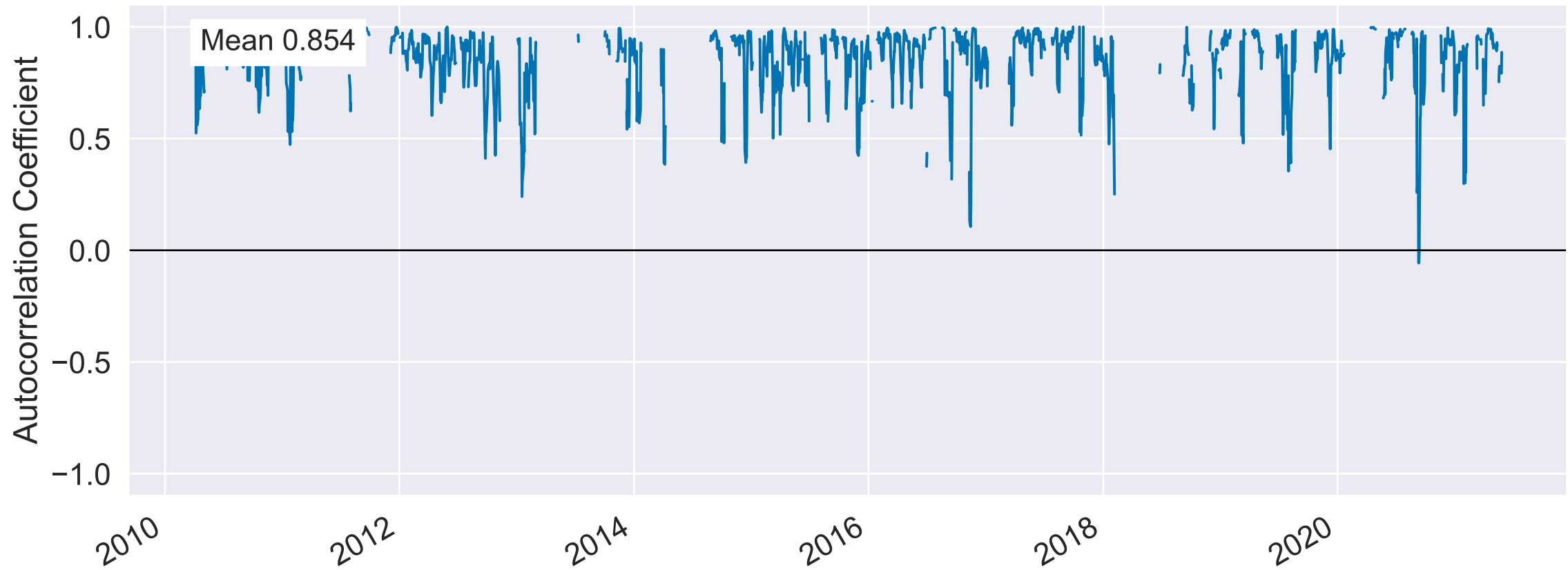
21D Period Top and Bottom Quantile Turnover



1D Period Factor Rank Autocorrelation



5D Period Factor Rank Autocorrelation



21D Period Factor Rank Autocorrelation

